

TIRTHARAJ DUTTA ROY

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SKILLS

- **Languages:** C++, Python, Java, JavaScript
- **Frameworks:** Flask, Flutter, Scikit-learn, XGBoost, MySQL
- **Tools:** Git, GitHub, VSCode, JupyterLabs, Excel
- **Specializations:** Data Processing, Model Building and Evaluation, ML Algorithms, NumPy, Pandas

PROJECTS

- **Student-Dropout Predictor** Nov'25 – Dec'25
 - Built a classification model to predict academic failure using a dataset of **3,000+ student records**, featuring **15+** demographic and academic indicators.
 - Handled class imbalance (only **12%** dropout rate) by applying **SMOTE (Synthetic Minority Oversampling Technique)**, which boosted the minority class **Recall score from 0.78 to 0.95**
 - Deployed via **Streamlit**, enabling non-technical staff to input data and generate risk probability scores with **<500ms inference time**.

Tech: Python, Streamlit, Scikit-Learn, Pandas, SMOTE, GitHub, Git.
- **Image to PDF Converter** Jul'25 – Aug'25
 - Developed an offline-first mobile application optimized for high-concurrency batch processing of **50+ high-resolution images**.
 - Designed the architecture to run **100% locally**, eliminating cloud latency and ensuring absolute user privacy with **zero external API dependencies**.
 - Implemented file management using dart:io, allowing users to rename files and share documents **instantly (<1s)** via the native OS share sheet.

Tech: Flutter, Dart, Android Studio, path_provider, share_plus, GitHub, Git.
- **Emotion-Based Poem Generator** Jun'25 – Jul'25
 - Engineered a web application utilizing the OpenAI API to generate emotion-based poetry, designing a **single, context-aware system prompt** capable of producing culturally nuanced output in **3+ languages** (e.g., English, Hindi, Spanish).
 - Enforced strict prompt constraints to prevent generic rhyming, ensuring the model prioritizes abstract imagery and maintains **tonal consistency** regardless of the input language.
 - Leveraged asynchronous JavaScript (Fetch API) to manage **3-4 second API response times**, implementing a skeleton loader to ensure **60 FPS UI smoothness** during generation.

Tech: HTML5, CSS3, JavaScript, OpenAI API, Fetch API, GitHub, Git.

TRAINING

- AI-Powered NLP Course – AlgoTutors Jun'25 – Aug'25
- Artificial Intelligence Course – Teachnook Nov'23 – Dec'23

CERTIFICATES

- ChatGPT Made Easy: AI Essentials for Beginners - Udemy Aug'25
- Course completion on Data Structures and Algorithms - iamneo Dec'24
- Fundamentals of Network Communication – Coursera Sep'24
- The Complete Python Bootcamp from Zero to Hero in Python - Udemy Dec'23

ACHIEVEMENTS

- **Social Worker in local NGO – SURJUDAYA**
Showcased teamwork skills along with meaningful Community Work
- **3rd Place: University-wide Group Discussion Competition**
Ranked 3rd out of 30+ participants in a university-wide Group Discussion event organized by the **Centre of Professional Enhancement (CPE)**
- **Finalist: Binary Blitz Hackathon**
Demonstrated Web-Development skills among hundreds of participants and reached the final round
- **SIH 2024 University Preliminary Hackathon**
Selected to compete in the University-level preliminary round, validating my ability to **analyze problem statements** and propose feasible technical solutions.

EDUCATION

- **Lovely Professional University** Punjab, India
Bachelor of Technology – Computer Science and Engineering August 2023 - Present
- **DPS, Digboi** Assam, India
Intermediate April 2021 – March 2022
- **Shiksha – The Gurukul** Assam, India
Matriculation April 2019 – March 2020